EECS 388

Introduction to Computer Security

Lecture 1:

Welcome!
The Security Mindset

September 1, 2026

Prof. Halderman and Prof. Ensafi



Today's Lecture

Welcome everyone,
both in the classroom and participating remotely!

Today, we'll cover:

- Who we are
- Goals for the course
- The security mindset
  - Thinking like an attacker
  - Thinking as a defender
- Course mechanics
- Security ethics



Who Are We?

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Our Outstanding TAs



Charlie Herz



Khush Bafna



Joy Han



Michael Jiang



Ella Mendelson

Course Topics

EECS 388 teaches the **security mindset**, and the principles and practices of security. Learn foundations of building, using, and managing secure systems.

1. **The Security Mindset**	threat modeling, thinking like an attacker
2. **Applied Cryptography**	randomness, ciphers and MACs, public key cryptography, secure channels
3. **Network Security**	web security, TLS and HTTPS, authentication, network attacks and defenses
4. **Host/Application Security**	control hijacking, malware, isolation, side channels, AI security, digital forensics
5. **Privacy/Security in Context**	privacy, anonymity, election security, Internet censorship, physical security

Learning Objectives

Critical thinking:

- How to think like an attacker, anticipate threats
- How to reason about threats and risks
- How to balance security costs and benefits

Technical skills:

- Know how to better protect yourself
- Understand common vulnerabilities and defensive tools
- Basics of building and managing systems more securely

Greater understanding of concepts from all layers of the computing stack

~~Learn to be a 1337 hax0r.~~ Learn to be an ethical, security-conscious citizen with the **humility** necessary to engineer secure systems

What is computer security?

Security

Difference between these disasters?





Meet the Adversary

a.k.a. **the attacker**

Security studies how systems behave in the presence of an adversary.

An *intelligence* that actively tries to cause the system to misbehave.

Can be a person, group, institution, or AI.

Often talked about in the abstract.

"Know your enemy." —Sun Tzu, *The Art of War*.

Movation? Capabilities? Opportunity?



The Security Mindset

The **security mindset** encourages thinking about **how attackers could cause systems to fail**, in order to head off problems before they are exploited.

Thinking like an attacker

- Understand techniques for circumventing security.
- Look for ways security can break, not excuses why it won't.

  Always ask: **"What could go wrong?"**

Thinking as a defender

- Know what you're defending, and against whom.
- Weigh benefits vs. costs: No system ever completely secure.

  Essentially "rational paranoia"

Thinking Like an Attacker

Imagine you are trying to attack the system:

- Look for **weakest links** (easiest places to attack)
- Identify **assumptions** security depends on. Are any of them false?
  example: "executing a program incorrectly and cleaning up is the same as not executing it" (led to Spectre and Meltdown)
- Think **outside the box**: Attackers not constrained by system designers' skill sets.
  example: software could be attacked via hardware vulnerabilities or physical security weaknesses

Helps to have a broad understanding of all the parts of how computer systems work.

Practice thinking like an attacker:

For every system you encounter, consider what it means for it to be secure, and image how attackers could exploit it.

Thinking like an attacker *shouldn't be* taboo… just don't follow through on your thinking!



Thinking Like an Attacker

Thought exercise:

The Beyster Building was designed with locks so that only those with a valid key (MCard) can enter.

How to break in to the Beyster Building?

(Remember: Weakest links? Security assumptions? Thinking outside the box?)

Thinking Like an Attacker

Thought exercise:

Other security systems you interact with in everyday life?

What could go wrong?

Get in the habit of practicing adversarial thinking.
For every system you encounter, ask yourself how attackers could exploit it.

Thinking as a Defender



Security policies

- What **assets** are we trying to protect?
- What **properties** are we trying to enforce?
  e.g.: **confidentiality**, **integrity**, **availability**, **authenticity**

Challenge is to think rationally and rigorously about potential risks.

Rational paranoia

Threat modeling

- Who are we defending against? Attacker motivations?
- Which threats should we disregard? ("Outside our threat model")

Assessing risk

- What would breaches cost us? Money, safety, reputation, … How likely? How to recover?

Selecting countermeasures

- Costs vs. benefits? No security mechanism is free.
  Indirect costs include added complexity, false positives, lost productivity …
- Technical or nontechnical approaches e.g., law enforcement, procedures, training

Thinking as a Defender

Thought exercise:

Defending your apartment or dorm room?

Think in terms of:

- Assets?
- Adversaries?
- Likelihood?
- Countermeasures?
- Costs/benefits?

Thinking as a Defender

Thought exercise (later, on your own):

Using a credit card safely, as a consumer?

Think in terms of:

- Assets?
- Adversaries?
- Likelihood?
- Countermeasures?
- Costs/benefits?

Secure Design and Assessment

Security is a **process** (something you do), not a product (something you buy)

It's not easily retrofitted—must be practiced continuously during development.

Good design strives for **defense-in-depth**

Layered defenses, so that multiple failures are necessary for an attack to succeed.

Common analyst mistake: Trying to convince yourself the system is secure

Better approach: Identify weaknesses in the design and mitigate them

Where to focus attention when analyzing a system's scrutiny:

- **Attack surface** (parts of the system **exposed to attackers**)
- **Trusted components** (parts that **must function correctly** for system to be secure)

Recall Our Learning Objectives ...

Critical thinking:

- How to think like an attacker, anticipate threats
- How to reason about threats and risks
- How to balance security costs and benefits

Technical skills:

- Know how to better protect yourself
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- Basics of building and managing systems more securely

Greater understanding of concepts from all layers of the computing stack

~~Learn to be a 1337 hax0r.~~ Learn to be an ethical, security-conscious citizen with the **humility** necessary to engineer secure systems

Lecture and Lab Mechanics

Lectures: Cover security concepts and their application

Encouraged: Attend lectures in person

Everyone is welcome, but seats only reserved for those in Section 001

Permitted: Watch lecture video asynchronously

Either way: Complete a **brief quiz** on Canvas before the next lecture day

Labs: Cover vital tools and techniques for completing the projects

Encouraged: Attend lab in person

May attend any section, but seats only reserved for your registered lab section

Permitted: Watch lab video asynchronously

First labs meet Fri./Mon. after Labor Day. Videos will be posted each week

No capacity limit: Everyone eligible has permission to enroll

Attendance

Choose the mode of learning that works best for you

Keep up with the material by attending in person and/or reviewing materials online

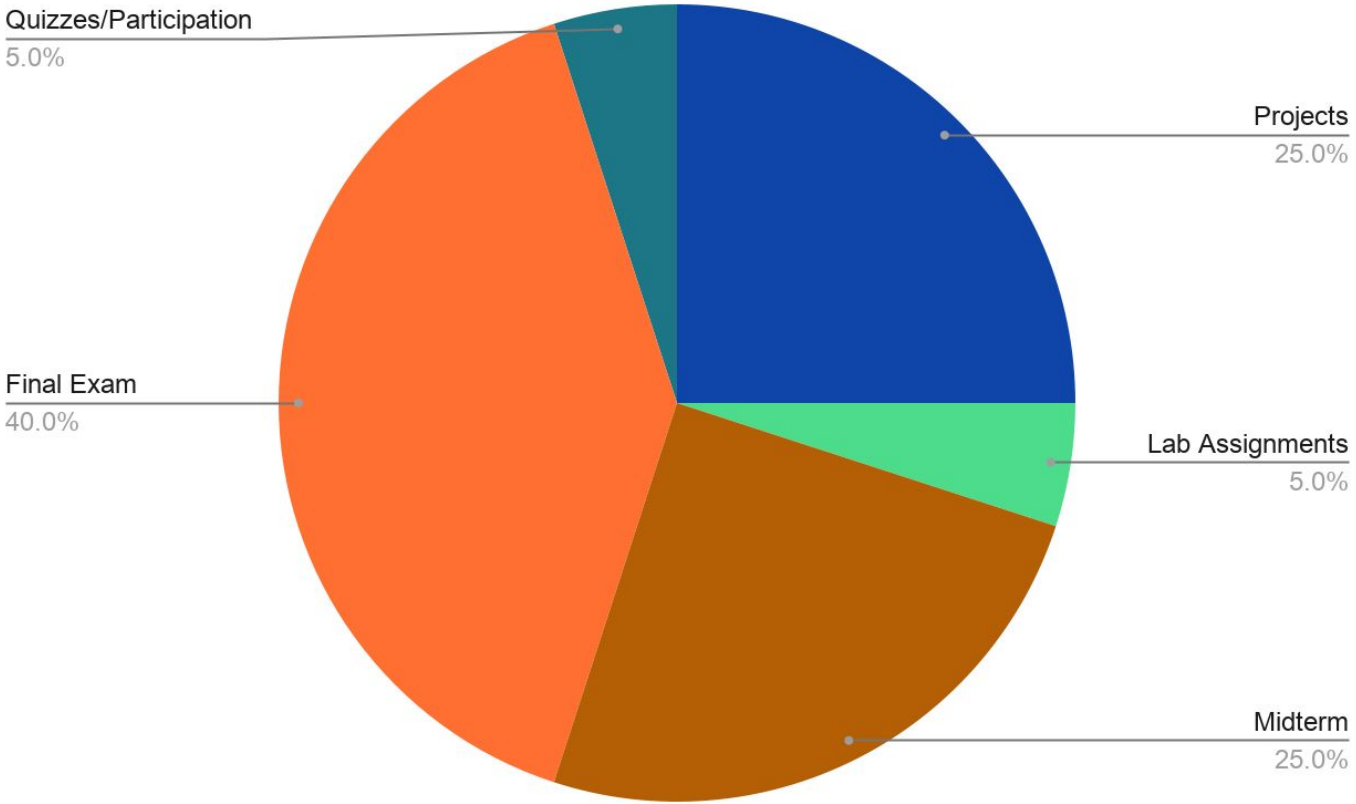
Default policy: We encourage coming to class, but we won't make you

Optional policy: Strict attendance with penalties (a *Ulysses pact*)

If you think you'd benefit from lecture but *know* you won't show up unless we make you, you can **opt-in** to this strict attendance policy. Email EECS388-staff@umich.edu to sign up.

- If you opt for strict attendance, your decision will be irrevocable
- We'll track your attendance with a sign-in sheet at the front of the lecture room
- You'll be allowed to miss up to two lectures without penalty
- After that, we'll reduce your overall course grade by 0.5% for every time you fail to sign the sheet, up to a maximum of 5%
- Absences will be excused only under extraordinary circumstances with documentation

Grading


Quizzes/Participation
5.0%
Projects
25.0%
Final Exam
40.0%
Lab Assignments
5.0%
Midterm
25.0%

Participation and Quizzes (5%)

2.5%

Intellectual Contributions

in class, OH, or Piazza

Goal: Encourage you to be a positive part of the learning experience. E.g.:

- asking good questions
- making thoughtful comments
- helping others appropriately

We look for quality over quantity and a high signal-to-noise ratio.



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+

2.5%

Brief Quizzes on Canvas

after each lecture

(due before day of next lecture)

Goal: Encourage you to keep up with the lectures

First Lecture Quiz
Available today
Due by **midnight tomorrow**

Projects (25%)

Five technical projects

1. Cryptography
2. Web security
3. Network security
4. Application security
5. Computer forensics

Project 1: Cryptography
Available Thursday

Part 1 due **Sept. 17** at 6 p.m.
Part 2 due **Sept. 24** at 6 p.m.

May work individually or with a partner for Projects 1–4; **must** work with a partner for Project 5

Upper-level course projects often require learning on your own, building on what we cover in class. Make sure you follow lecture and lab, and start early.

Projects will be distributed on EECS388.org and submitted via Autograder.io

Lab Assignments (5%)

Simple labs accompanying the projects

Interactive introductions to languages or tools

Must be completed individually

Lab 1: Docker and Python
Available Thursday

Due **Sept. 10** at 6 p.m

Upper-level course projects often require learning on your own, building on what we cover in class. Make sure you follow lecture and lab, and start early.

Lab assignments will be distributed on EECS388.org and submitted via Autograder.io

Assignment Policies

Lateness*

We want work in promptly so we can move on and review solutions together.

Labs and Projects:

- **3 late days** for all routine illness, tech problems, scheduling conflicts, etc.

Quizzes:

- **Half credit** if in before final exam day

Exceptions only for ***extreme*** medical or personal emergencies, with documentation

*See policy details on EECS388.org

Please start assigned work early.

Collaboration

We encourage you to learn together by discussing course **concepts.**

No cheating policy:

- Don't give, ask for, or accept **hints/solutions**, including from AI
- Don't copy others' work
- Don't share or post your solutions

Questions? Ask us

We have **booby traps** in place to catch cheaters. We'll report you to the Honor Council after the final exam.

Generative AI Policy

The point of this class is to teach *you* to solve certain problems *yourself*.

Prohibited:

Using AI for the central intellectual work (the *crux*) of an assignment.

Don't use AI to solve/get hints about parts of the assignment that develop security-related knowledge or skills.

Ask, ***"Would using AI for this bypass what the assignment is trying to teach me?"***

WARNING The exams test skills you develop by completing the projects. If you cheat and have AI do the projects, expect a C or lower on the exams.

Permitted:

Using AI for *secondary tasks* so you can focus on project learning objectives.

- clarifying terms or background concepts
- setting up the project environment
- helping with syntax, APIs, or libraries
- code unrelated to the crux of the project
- improving clarity/organization of code

Projects will ship will an `AGENTS.md` file as a guardrail to help AI help you without spoiling the learning experience. However, responsibility for following the rules rests with you!

Midterm (25%) and Final (40%)

Midterm Exam

Wednesday, October 21, 7–9 p.m. IN PERSON
Covers first half of the course

Final Exam

Friday, December 18, 7–9 p.m. IN PERSON
Covers the entire course

Both exams are IN PERSON only. Plan travel accordingly.

Exams will test concepts and techniques that you learn from **lectures *and* projects**.

We'll provide practice exams and hold review sessions prior to each test.

If you require testing accommodations, upload SSD form via umich-accommodate.symplicity.com.



Communication and Resources

Website: EECS388.org
overview, schedule, assignments

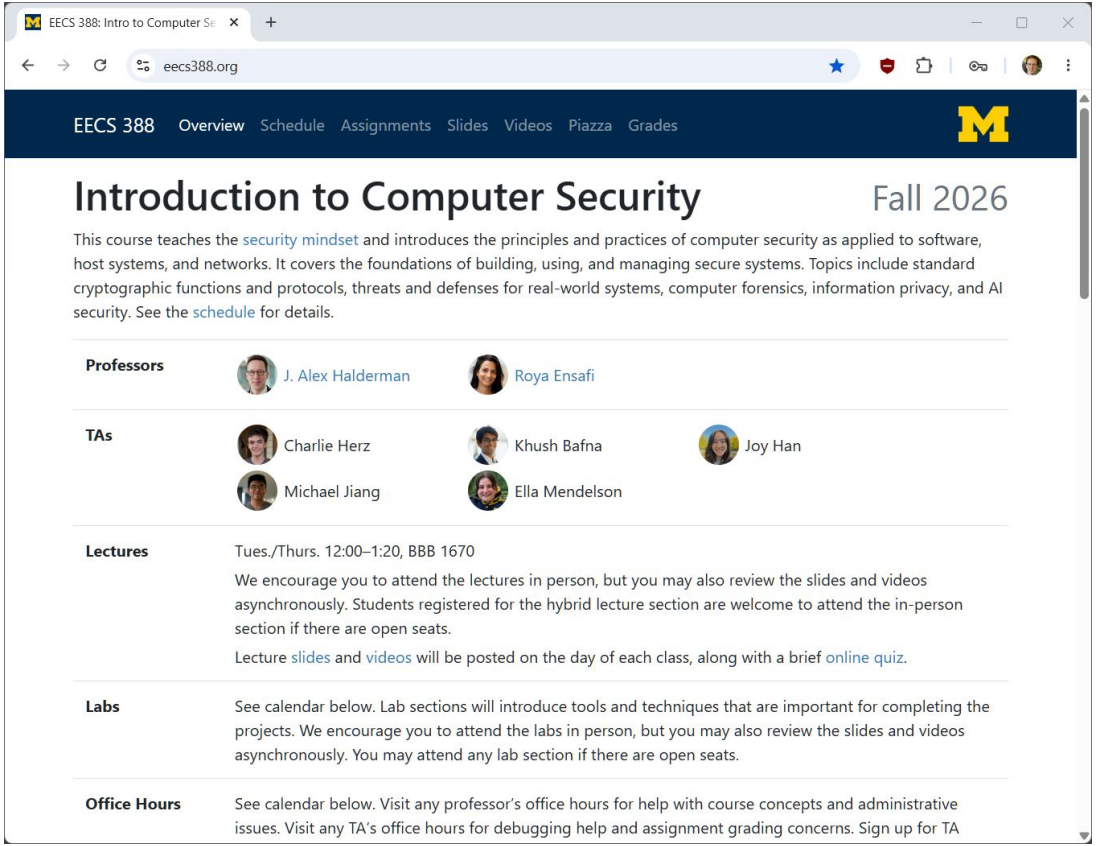
Piazza
questions, discussion, slides, announcements

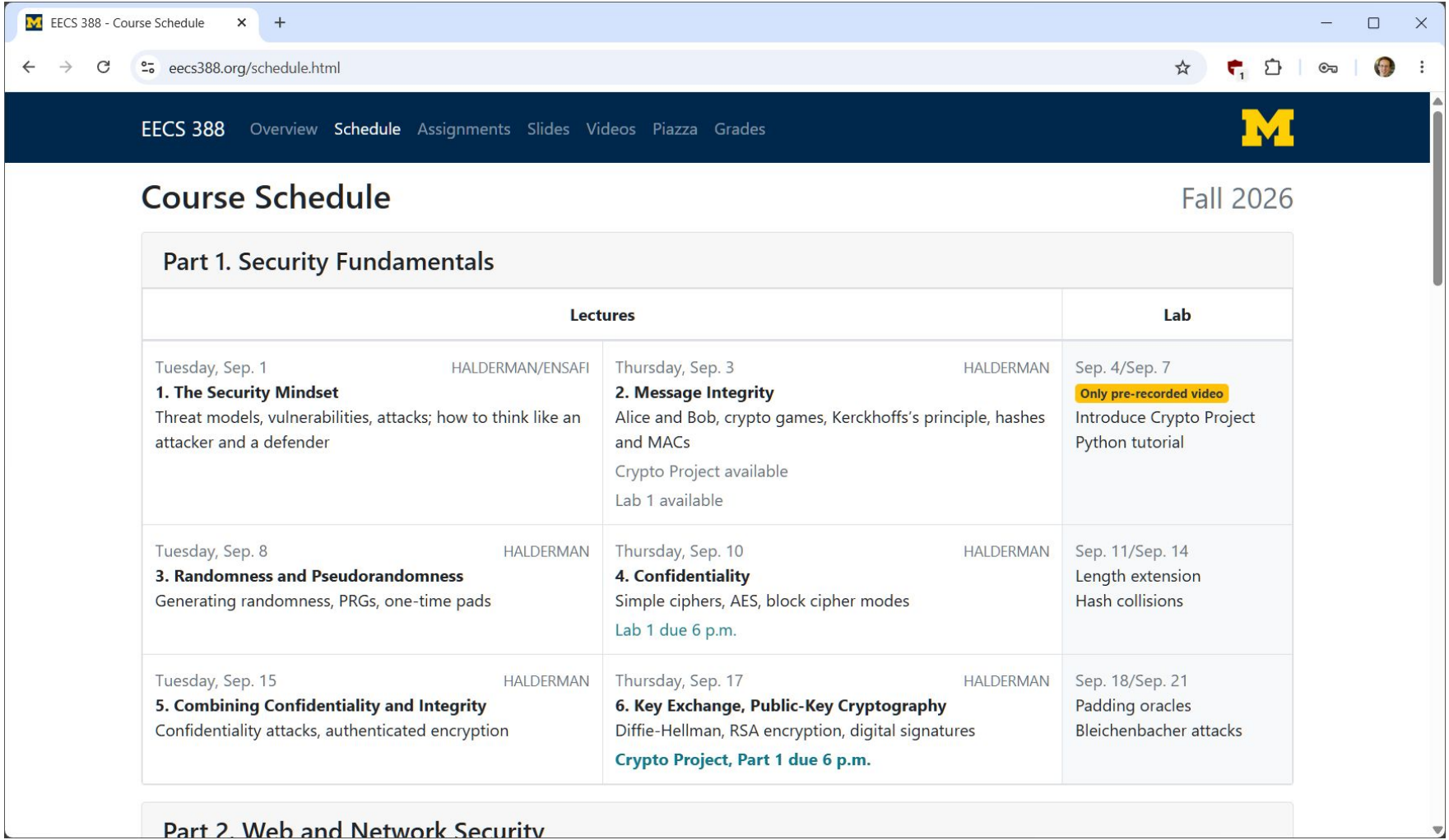
Canvas
gradebook, lecture videos

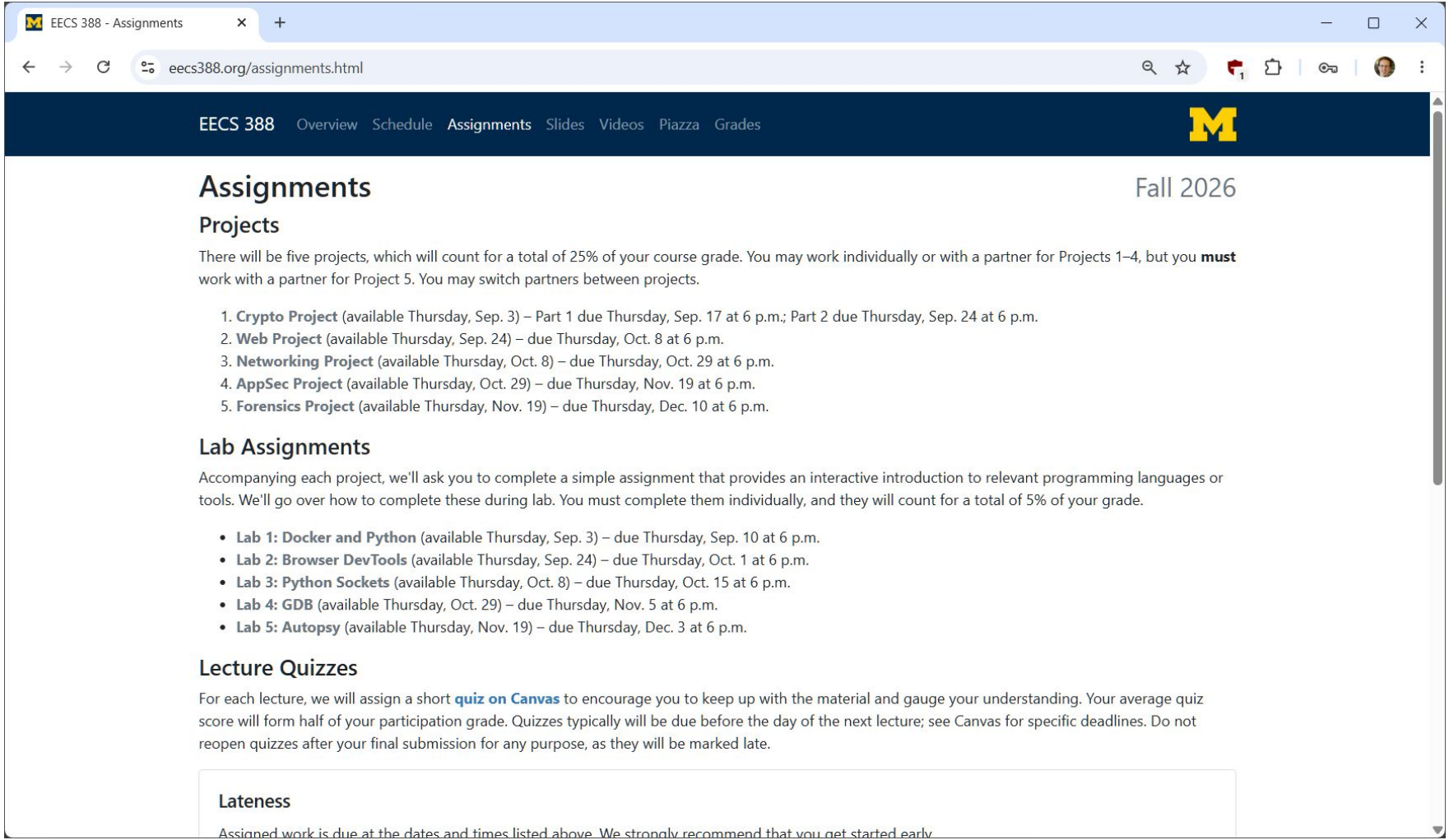
Autograder.io
assignment submission/grading

Office Hours
(see calendar on course website)

EECS388-staff@umich.edu
administrative issues




EECS 388 - Course Schedule
eecs388.org/schedule.html
EECS 388 Overview Schedule Assignments Slides Videos Piazza Grades
Course Schedule
Fall 2026
Part 1. Security Fundamentals
Lectures
Lab
Tuesday, Sep. 1 HALDERMAN/ENSAFI
1. The Security Mindset
Threat models, vulnerabilities, attacks; how to think like an attacker and a defender
Thursday, Sep. 3 HALDERMAN
2. Message Integrity
Alice and Bob, crypto games, Kerckhoffs's principle, hashes and MACs
Crypto Project available
Lab 1 available
Sep. 4/Sep. 7
Only pre-recorded video
Introduce Crypto Project
Python tutorial
Tuesday, Sep. 8 HALDERMAN
3. Randomness and Pseudorandomness
Generating randomness, PRGs, one-time pads
Thursday, Sep. 10 HALDERMAN
4. Confidentiality
Simple ciphers, AES, block cipher modes
Lab 1 due 6 p.m.
Sep. 11/Sep. 14
Length extension
Hash collisions
Tuesday, Sep. 15 HALDERMAN
5. Combining Confidentiality and Integrity
Confidentiality attacks, authenticated encryption
Thursday, Sep. 17 HALDERMAN
6. Key Exchange, Public-Key Cryptography
Diffie-Hellman, RSA encryption, digital signatures
Crypto Project, Part 1 due 6 p.m.
Sep. 18/Sep. 21
Padding oracles
Bleichenbacher attacks
Part 2. Web and Network Security


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Assignments
Fall 2026
Projects
There will be five projects, which will count for a total of 25% of your course grade. You may work individually or with a partner for Projects 1–4, but you must work with a partner for Project 5. You may switch partners between projects.
1. Crypto Project (available Thursday, Sep. 3) – Part 1 due Thursday, Sep. 17 at 6 p.m.; Part 2 due Thursday, Sep. 24 at 6 p.m.
2. Web Project (available Thursday, Sep. 24) – due Thursday, Oct. 8 at 6 p.m.
3. Networking Project (available Thursday, Oct. 8) – due Thursday, Oct. 29 at 6 p.m.
4. AppSec Project (available Thursday, Oct. 29) – due Thursday, Nov. 19 at 6 p.m.
5. Forensics Project (available Thursday, Nov. 19) – due Thursday, Dec. 10 at 6 p.m.
Lab Assignments
Accompanying each project, we'll ask you to complete a simple assignment that provides an interactive introduction to relevant programming languages or tools. We'll go over how to complete these during lab. You must complete them individually, and they will count for a total of 5% of your grade.
Lab 1: Docker and Python (available Thursday, Sep. 3) – due Thursday, Sep. 10 at 6 p.m.
Lab 2: Browser DevTools (available Thursday, Sep. 24) – due Thursday, Oct. 1 at 6 p.m.
Lab 3: Python Sockets (available Thursday, Oct. 8) – due Thursday, Oct. 15 at 6 p.m.
Lab 4: GDB (available Thursday, Oct. 29) – due Thursday, Nov. 5 at 6 p.m.
Lab 5: Autopsy (available Thursday, Nov. 19) – due Thursday, Dec. 3 at 6 p.m.
Lecture Quizzes
For each lecture, we will assign a short quiz on Canvas to encourage you to keep up with the material and gauge your understanding. Your average quiz score will form half of your participation grade. Quizzes typically will be due before the day of the next lecture; see Canvas for specific deadlines. Do not reopen quizzes after your final submission for any purpose, as they will be marked late.
Lateness
Assigned work is due at the dates and times listed above. We strongly recommend that you get started early

Coming Up

Reminders:

Complete the Quiz on Canvas after every lecture, due before the day of the next lecture

Thursday

Intro to Cryptography

Alice and Bob
Kerckhoffs's principle
Hashes and MACs

Coming Weeks

Applied Cryptography

Pseudorandomness
Integrity and confidentiality
Public key cryptography